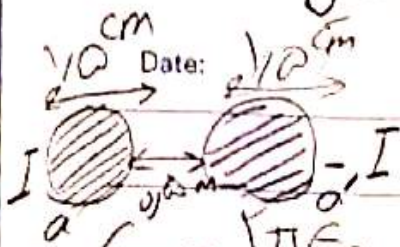


"آزمون بزرگری ۱۳۵۲/۱۰/۲۵" "تعمین سری چهارم کامل"



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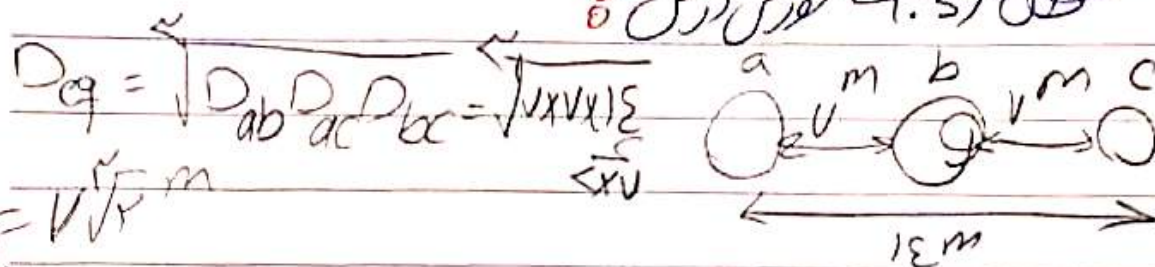
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مسئله 4.32 / هرکس درک

$$C_n = \frac{2\pi\epsilon_0}{\ln(D/r)} = \frac{2\pi \times 1.125 \times 10^{-12}}{\ln\left(\frac{0.15}{0.015}\right)} = 1.125 \times 10^{-11} \frac{F}{m}$$

$$Y_n = j\omega C_n = j(2\pi \times 40)(1.125 \times 10^{-11}) \frac{S}{m} \times \frac{10^3 m}{km} = j 2.825 \times 10^{-9} \frac{S}{km}$$

مسئله 4.39 / هرکس درک



$$D_{eq} = \sqrt{D_{ab} D_{ac} D_{bc}} = \sqrt{r \times 1.5 \times 1.5} = 1.5 \sqrt{r}$$

بجای ر استفاده می‌کنیم

$$GMR = 0.105 r^{0.775} ft \Rightarrow GMR = 0.105 r^{0.775} ft \times \frac{0.1508 m}{1 ft} = 1.27 r^{0.775} mm$$

$$C_1 = \frac{2\pi\epsilon_0}{\ln(D/GMR)} = \frac{2\pi \times 1.125 \times 10^{-12}}{\ln\left(\frac{1.5 \sqrt{r}}{0.0127 r^{0.775}}\right)} = 1.125 \times 10^{-11} \frac{F}{m}$$

$$Y_1 = j\omega C_1 = j \times 2\pi \times 90 \times 1.125 \times 10^{-11} \frac{S}{m} \times \frac{10^3 m}{km} = j 1.19 \times 10^{-9} \frac{S}{km}$$

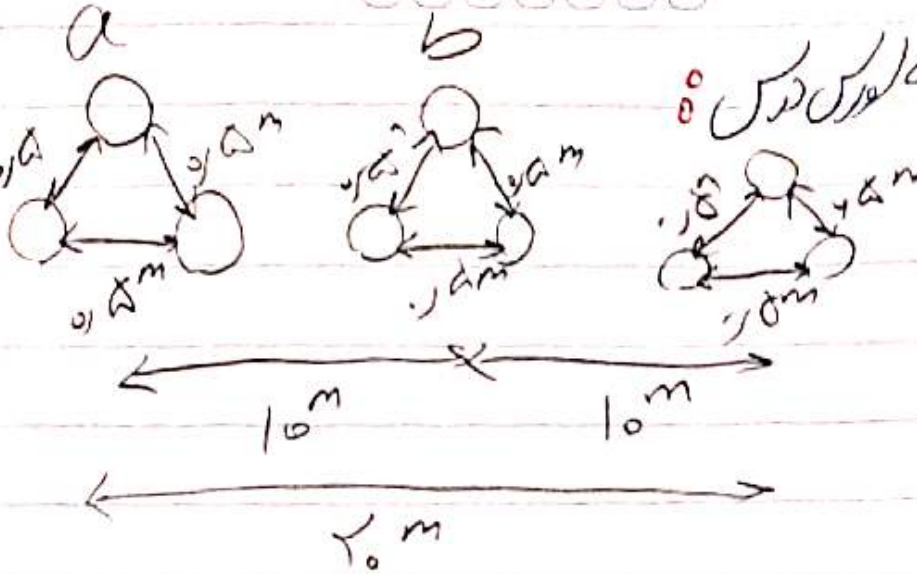
$$I_{chg} = \sqrt{19} W = (\sqrt{1.19 \times 10^{-9}} \times 110) \left(\frac{220}{\sqrt{3}} \right) = 5.1294 \times 10^{-2} \frac{KA}{PHASE}$$

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9.41

$$D_{eq} = \sqrt{D_{ab} D_{ac} D_{bc}} = \sqrt{10 \times 10 \times 10} = 10\sqrt{3}$$

از جدول $\rightarrow GMR_{a1} = 0.054 \text{ ft} = 0.054 \text{ ft} \times \frac{0.3048 \text{ m}}{1 \text{ ft}} = 0.0164 \text{ m}$

$$\Rightarrow GMR_a = \sqrt{GMR_{a1} \times r'} = \sqrt{0.0164 \times 0.02} = 0.0189 \text{ m}$$

$$C_1 = \frac{\gamma \pi \epsilon_0}{\ln\left(\frac{D_{eq}}{GMR_a}\right)} = \frac{\gamma \pi \times 8.854 \times 10^{-12}}{\ln\left(\frac{10\sqrt{3}}{0.0189}\right)} = 1.5 \times 10^{-11} \frac{\text{F}}{\text{m}}$$

$$Y_1 = j\omega C_1 = j \times 2\pi \times (60) \times 1.5 \times 10^{-11} \frac{\text{S}}{\text{m}} \times \frac{10 \text{ km}}{1 \text{ km}} = 5.65 \times 10^{-9} \frac{\text{S}}{\text{km}}$$

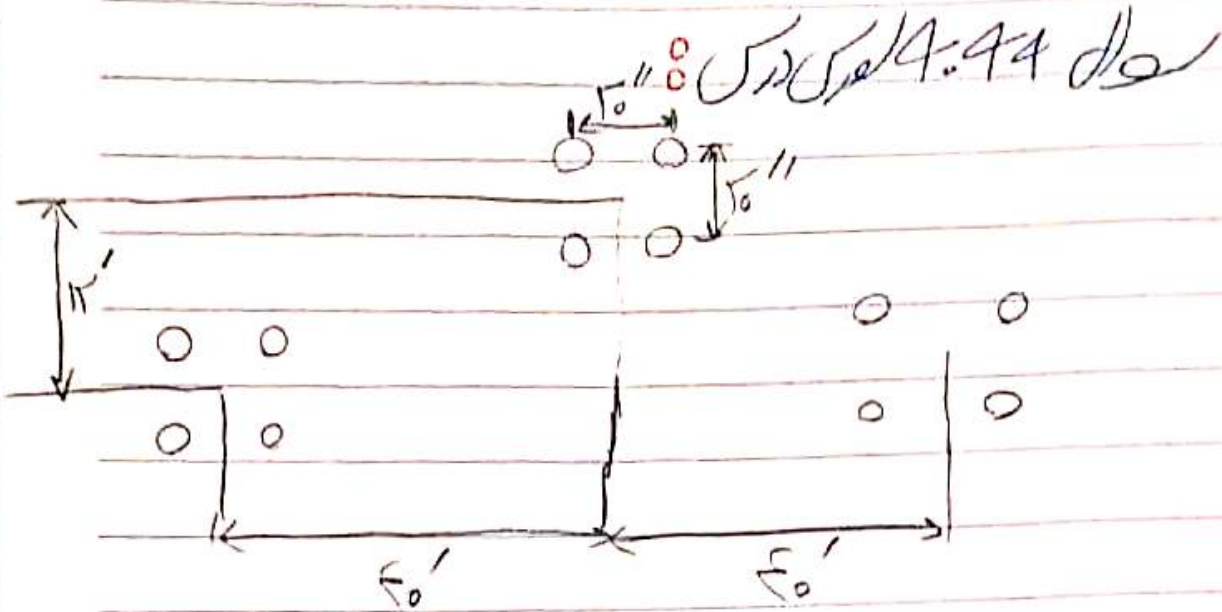
$$Q_1 = 225 Y_1 = (225) \times 5.65 \times 10^{-9} = 1.27 \frac{\text{MVAR}}{\text{km}}$$

1.27

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$$D_{11} = \sqrt{r_0' + r_0'} = \sqrt{100 + 188} = \sqrt{288} \approx 17.14 \text{ ft}$$

$$D_{22} = \sqrt{r_0' + r_0'} = \sqrt{100 + 188} = \sqrt{288} \approx 17.14 \text{ ft}$$

$$D_{12} = r_0' + r_0' = 10 \text{ ft} \Rightarrow D_{eq} = \sqrt{D_{11} D_{22} D_{12}}$$

$$= \sqrt{10 \times 17.14 \times 17.14} = \sqrt{2982} \approx 54.61 \text{ ft}$$

دو 4.44 کی دس $d = 1.194 \text{ in} \Rightarrow r = \frac{1.194}{2} = 0.597 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}}$

$$= 0.04975 \text{ ft} / d = 10 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}} = 1.44 \text{ ft}$$

$$GMR = 109 \times \sqrt{0.04975 \times 1.44} \approx 1.44 \text{ ft}$$

$$= 0.494 \text{ ft} \rightarrow C = \frac{2\pi\epsilon_0}{\ln\left(\frac{D_{eq}}{GMR}\right)} = \frac{2\pi \times 10^{-9}}{\ln\left(\frac{54.61}{0.494}\right)}$$

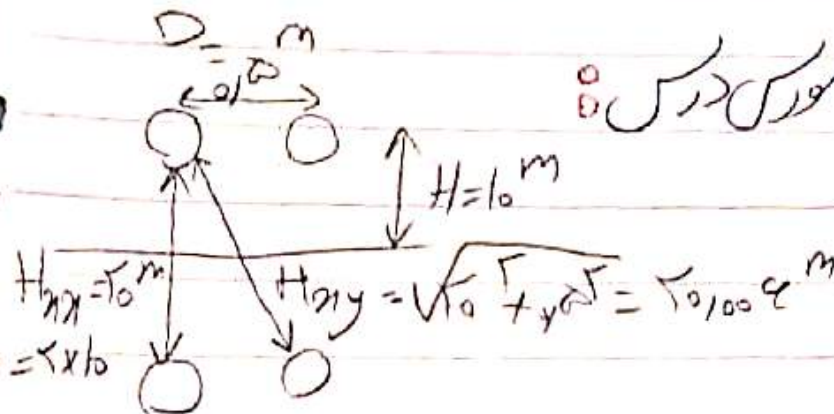
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$$= 15/11914 \times 10^{-12} \frac{F}{m} \times \frac{1909188 \text{ m}}{1 \text{ m}^2} = 2.105 \times 10^{-11} \frac{F}{\text{m}^2} \Rightarrow X_C = \frac{1}{\omega C}$$

$$= \frac{1}{2\pi f C} = \frac{1}{2\pi \times 40 \times 2.105 \times 10^{-11}} = 187.1 \text{ } \Omega \cdot \text{m}^2 \text{ Phase}$$



والد 4.45 درس

$$C_{mn} = \frac{2\pi \epsilon_0}{\ln\left(\frac{D}{r}\right) - \ln\left(\frac{H_{xy}}{H_{mn}}\right)} = \frac{2\pi \times 11.126 \times 10^{-12}}{\ln\left(\frac{0.5}{0.016}\right) - \ln\left(\frac{7.07}{10}\right)}$$

$$= 1.52 \times 10^{-11} \frac{F}{m}$$

خط طولی و عرضی از زمین
نسبت: 4.32
نسبت: 0.11

والد 4.46 درس

$$D_{eq} = \sqrt{12 \times K \times \frac{r}{15 \times 2}} = 12.52 / d = 0.228 \text{ m} \Rightarrow r = \frac{d}{2} = 0.114 \text{ m}$$

$$\Rightarrow C_{on} = \frac{2\pi \times 11.126 \times 10^{-12}}{\ln\left(\frac{12.52}{0.114}\right)} \Rightarrow X_C = \frac{1}{\omega C_{on}} = \frac{1}{2\pi \times 40} \times \frac{\ln\left(\frac{12.52}{0.114}\right)}{2\pi \times 11.126 \times 10^{-12}}$$

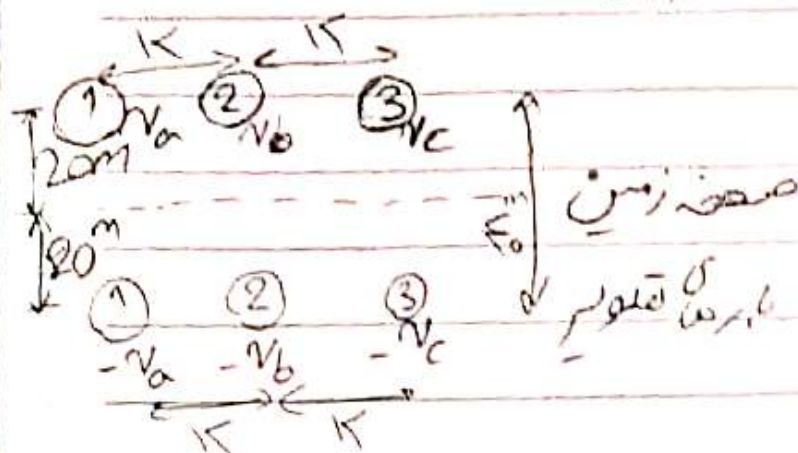
187.1 Ω.m

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$$\Rightarrow \text{For } 125 \text{ km} : X_c = \frac{2.1 \times 10^8}{1.5 \times 10^5} = 1400 \Omega$$



$$H_1 = H_2 = H_3 = 50 \text{ m}$$

$$H_{12} = H_{23} = \sqrt{50^2 + 12^2} = 51.491 \text{ m}$$

$$H_{13} = \sqrt{50^2 + 24^2} = 54.454 \text{ m}$$

$$D_{eq} = \sqrt[3]{12 \times 12 \times 24} = 15.118 \text{ m} / d = 0.054 \text{ m} \rightarrow r = \frac{d}{2} = 0.027 \text{ m}$$

$$\Rightarrow C = \frac{2\pi\epsilon_0}{\ln \frac{D_{eq}}{r} - \ln \frac{H_m}{H_s}} = \frac{2\pi \times 8.854 \times 10^{-12}}{\ln \left(\frac{15.118}{0.027} \right) - \ln \left(\frac{51.491}{50} \right)}$$

$$H_m = \sqrt[3]{H_{12} H_{23} H_{13}} = \sqrt[3]{51.491 \times 51.491 \times 54.454} = 52.55 \text{ m}$$

$$H_s = \sqrt[3]{H_1 H_2 H_3} = \sqrt[3]{50 \times 50 \times 50} = 50 \text{ m}$$

$$= 1.185 \times 10^{-9} \frac{\text{F}}{\text{m}} \Rightarrow X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 1.185 \times 10^{-9}} = 664.4 \Omega$$

$$\Rightarrow X_c = \frac{2.1 \times 10^8}{1.5 \times 10^5} = 1400 \Omega$$

4.50 d

$$C_{ph} = 1.185 \times 10^{-9} \frac{\text{F}}{\text{m}} \Rightarrow C_{py} = \frac{1}{3} C_{ph} = \frac{1}{3} \times 1.185 \times 10^{-9} = 3.95 \times 10^{-10} \frac{\text{F}}{\text{m}} \Rightarrow V_{py} = 50 \text{ kV}$$

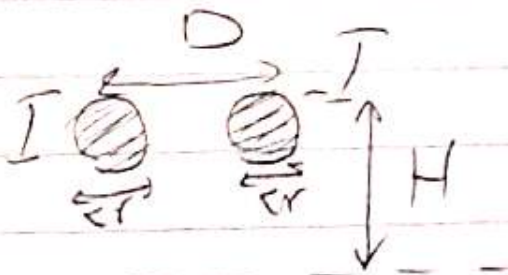
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$$q_x = C_{xy} V_{xy} = (4.15 \times 10^{-15} \frac{F}{m}) (50 \times 10^5 \frac{V}{m}) = 1.5854 \times 10^{-9} \frac{C}{m}$$

$$\Rightarrow E_T = \frac{q_x}{\pi \epsilon_0 r} = \frac{1.5854 \times 10^{-9}}{\pi \times 1.12 \times 10^{-5} \times 0.0005} = 5.114 \times 10^6 \frac{V}{m} = 5.114 \text{ kV/cm}$$

$$E = \frac{1.5854 \times 10^{-9}}{\pi \times 1.12 \times 10^{-5} \times \epsilon_0} \left(\frac{r_0}{r} - \frac{r_0}{r_0 + a} \right) = 1.5854 \times 10^6 \times \frac{1}{1.522} = 0.104508 \text{ kV/m}$$



و د پ د

$$C' = \frac{\pi \epsilon_0}{\ln\left(\frac{D}{r} \times \frac{2H}{\sqrt{D^2 + 4H^2}}\right)} = \frac{\pi \epsilon_0}{\ln\left(\frac{D}{r} \frac{1}{\sqrt{1 + \left(\frac{D}{2H}\right)^2}}\right)}$$

باردار لږ تر لږه
اثر زمین

$$C = \frac{\pi \epsilon_0}{\ln\left(\frac{D}{r}\right)}$$

د د باردار لږ تر لږه
اثر زمین

$$E_{\text{isol}} = \frac{C' - C}{C'} = 1 - \frac{C}{C'} = 1 - \frac{\ln\left(\frac{D}{r} \frac{1}{\sqrt{1 + \left(\frac{D}{2H}\right)^2}}\right)}{\ln\left(\frac{D}{r}\right)} = 1 - \frac{\ln\left(\frac{D}{r}\right) + \ln\left(\frac{1}{\sqrt{1 + \left(\frac{D}{2H}\right)^2}}\right)}{\ln\left(\frac{D}{r}\right)}$$

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$$\ln \frac{1}{\sqrt{1 + \left(\frac{D}{\sqrt{H}}\right)^2}}$$

$$= 1 - \left(1 + \frac{\ln\left(1 + \left(\frac{D}{\sqrt{H}}\right)^2\right)^{-\frac{1}{2}}}{\ln\left(\frac{D}{\sqrt{H}}\right)}\right)$$

$$= - \frac{-\frac{1}{2} \ln\left(1 + \left(\frac{D}{\sqrt{H}}\right)^2\right)}{\ln\left(\frac{D}{\sqrt{H}}\right)} = \frac{\ln\left(1 + \left(\frac{D}{\sqrt{H}}\right)^2\right)}{\ln\left(\frac{D}{\sqrt{H}}\right)} < \frac{\omega}{100}$$

$$\Rightarrow \ln\left(1 + \left(\frac{D}{\sqrt{H}}\right)^2\right) < \ln\left(\frac{D}{\sqrt{H}}\right)$$

$$\ln\left(\left(\frac{D}{\sqrt{H}}\right)^{0.1}\right) \Rightarrow$$

$$\Rightarrow 1 + \left(\frac{D}{\sqrt{H}}\right)^2 < \left(\frac{D}{\sqrt{H}}\right)^{0.1} \rightarrow \left(\frac{D}{\sqrt{H}}\right)^2 < \left(\frac{D}{\sqrt{H}}\right)^{0.1} - 1 \rightarrow H^2 > \frac{D^2}{\left[\left(\frac{D}{\sqrt{H}}\right)^{0.1} - 1\right]^2}$$

$$\Rightarrow H > \frac{D}{\sqrt{\left(\frac{D}{\sqrt{H}}\right)^{0.1} - 1}}$$